

UNIT – I : Object Oriented Programming

Basic concepts, Principles, Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments. Data Types, Variables, and Operators: Introduction, Data Types in Java, Declaration of Variables, Data Types, Constants. Introduction to Operators, Precedence and Associativity of Operators, Assignment Operator (=), Basic Arithmetic Operators, Increment (++) and Decrement (- -) Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators. Control Statements: Introduction, if Expression, Nested if Expressions, if- else Expressions, Ternary Operator?;, Switch Statement, Iteration Statements, while Expression, do-while Loop, for Loop, Nested for Loop, For-Each for Loop, Break Statement, Continue Statement.

// Average of three numbers Java Program

```
import java.util.*;
public class Average
{
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        float num1,num2,num3;
        System.out.println("Enter three float numbers");
        num1=s.nextFloat();
        num2=s.nextFloat();
        num3=s.nextFloat();
        float avg;
        avg=(num1+num2+num3)/3.0f;
        System.out.println("The avg is"+avg);
    }
}
```

javac Average.java

java -cp . Average

//Reading of two numbers and perform Addition

```
import java.util.*;
public class Read
{
    public static void main(String args[])
    {
```

```

        int a,b;
        System.out.println("Enter two values");
        Scanner s=new Scanner(System.in);
        a=s.nextInt();
        b=s.nextInt();
        System.out.println("The result of sum is"+(a+b));
    }
}

```

//First Java Program

```

import java.util.*;//package stmt
public class First //main class
{
    public static void main(String args[])//main method
    {
        System.out.println("Hello Open Elective Students");
    }
}

```

Command Line Argument:

```

import java.util.*;
public class CommandLine
{
    public static void main(String args[])
    {
        for(int i=0; i<args.length; i++)
        {
            System.out.println(args[i]);
        }
    }
}

```

/*

D:\Openelective2>javac CommandLine.java

D:\Openelective2>java -cp . CommandLine CIV EEE MECH ECE

CIV
EEE
MECH
ECE*/

Addition of two numbers (Command Line Argument):

```
import java.util.*;
public class CommandAdd
{
    public static void main(String args[])
    {
        int a= Integer.parseInt(args[0]);
        int b=Integer.parseInt(args[1]);

        System.out.println("the addition is" +(a+b));
    }
}
/*D:\Openelective2>javac CommandAdd.java
```

```
D:\Openelective2>java -cp . CommandAdd 10 20
the addition is30
*/
```

//java program to demonstrate operators

```
import java.util.*;
public class Operators
{
    public static void main(String args[])
    {
        int a=10,b=20;
        Scanner s=new Scanner(System.in);

        System.out.println("The modulus of a and b is" +(a%b)); //10
        System.out.println(" greater than equal to" +(a>=b));    //false
        System.out.println("logical and is" +(a > b && a++ < b++)); //false
        System.out.println("The value of a and b is" +a+"\t"+b);    //11 21
        System.out.println("The value of complement a is" +~a);    //-10
        a += 2;
```

```

        System.out.println("Is s is an instance of Scanner"+(s instanceof
Scanner));//true
        System.out.println("The value of a and b is"+a+"\t"+b);//13 21
    }
}

```

/*

D:\Openelective2>javac Operators.java

D:\Openelective2>java -cp . Operators

The modulus of a and b is10

greater than equal to false

logical and isfalse

The value of a and b is10 20

The value of complement a is-11

Is s is an instance of Scannertrue

The value of a and b is12 20

*/

//java program to demonstrate literals

```
import java.util.*;
```

```
public class Literals
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        int a=123,b=012,c=0x1A,d=0b1011;
```

```
        float e=23.34f;
```

```
        double f=23.4E+3;
```

```
        char g='@';
```

```
        String h="Gudlavalleru";
```

```
        boolean i=false;
```

```
        System.out.println("The value of a is "+a);
```

```
        System.out.println("Octal"+b+"\nHexa"+c+"\nbinary"+d);
```

```
        System.out.println("float"+e+"\nDouble"+f);
```

```
        System.out.println("Char"+g+"\n\u0041"+"n\u0042"+"n\u0043");
```

```
        System.out.println("String"+h+"\nBoolean"+i);
```

```
    }
```

```
}/*
```

```
D:\Openelective2>javac Literals.java
```

```
D:\Openelective2>java -cp . Literals
```

```
The value of a is 123
```

```
Octal10
```

```
Hexa26
```

```
binary11
```

```
float23.34
```

```
Double23400.0
```

```
Char@
```

```
A
```

```
BZ
```

```
StringGudlavalleru
```

```
Booleanfalse*/
```

Control Statements

Single way Selection Statement:

// java program to demonstrate person eligible for voting using **simple if**

```
import java.util.*;
public class SimpleIf
{
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        System.out.println("Enter the age of the person");
        int age= s.nextInt();
        if(age>17)
        {
            System.out.println("The person is eligible for vote");
        }
    }
}
```

Two way:

//java program to demonstrate smaller than two numbers using **if-else** statement

```
import java.util.*;
public class IfElse
{
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        System.out.println("Enter two numbers");
        int num1=s.nextInt();
        int num2=s.nextInt();
        if( num1< num2)
        {
            System.out.println(""+num1+" is small number");
        }
        else
        {
            System.out.println(""+num2+" is small number");
        }
    }
}
```

//java program to implement greatest among three numbers using **nested if**

```
import java.util.*;
```

```
public class NestedIf
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        Scanner s=new Scanner(System.in);
```

```
        System.out.println("Enter the three numbers");
```

```
        int a=s.nextInt();
```

```
        int b=s.nextInt();
```

```
        int c=s.nextInt();
```

```
        if(c>a)
```

```
        {
```

```
            if(c>b)
```

```
            {
```

```
                System.out.println("C is greater number");
```

```
            }
```

```
            else
```

```
            {
```

```
                System.out.println("B is greater number");
```

```
            }
```

```
        }
```

```
        else
```

```
        {
```

```
            if(b>a)
```

```
            {
```

```
                System.out.println("B is greater number");
```

```
            }
```

```
            else
```

```
            {
```

```
                System.out.println("A is greater number");
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

Multi way selection statement:

//java program to implement greatest among three using **Else-if Ladder**

```
import java.util.*;
```

```
public class Elself
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        Scanner s=new Scanner(System.in);
```

```
        System.out.println("Enter the three numbers");
```

```
        int a=s.nextInt();
```

```
        int b=s.nextInt();
```

```
        int c=s.nextInt();
```

```
        if(b>a&& b>c)
```

```
        {
```

```
            System.out.println("B is greater number");
```

```
        }
```

```
        else if(c>a)
```

```
        {
```

```
            System.out.println("C is greater number");
```

```
        }
```

```
        else
```

```
        {
```

```
            System.out.println("A is greater number");
```

```
        }
```

```
    }
```

```
}
```



```

//java program to demonstrate switch
import java.util.*;
public class Switch
{
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        System.out.println("Enter a and b values");
        int a=s.nextInt();
        int b=s.nextInt();

        System.out.println("Enter the menu");
        System.out.println("add. addition");
        System.out.println("sub. Substraction");
        System.out.println("mul. Multiplication");
        System.out.println("div. Division");
        System.out.println("enter the choice");

        String choice = s.next();
        switch(choice)
        {
            case "add":
                System.out.println("The addition is" +(a+b));
                break;
            case "sub":
                System.out.println("The sub is" +(a-b));
                break;
            case "mul":
                System.out.println("The mul is" +(a*b));
                break;
            case "div":
                System.out.println("The div is" +(a/b));
                break;
            default:
                System.out.println("Enter the valid choice");
        }
    }
}

```

Iterative Statements:

//java program to print sum of n natural numbers using **while loop**

```
import java.util.*;
public class While
{
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        System.out.println("Enter the value of n");
        int n=s.nextInt();
        int i=1,sum=0; //initialization
        while(i<=n)
        {
            sum=sum+i;
            i++;
        }
        System.out.println("The sum of n natural numbers is"+sum);
    }
}
```

//Java program to print factorial of given number using **for loop**

```
import java.util.*;
public class For
{
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        System.out.println("Enter the value of n");
        int n=s.nextInt();
        int fact=1;
        for(int i=n;i>=1;i--)
        {
            fact=fact*i;
        }
        System.out.println("the factorial of given number is"+fact);
    }
}
```

```
//java program to demonstrate nested loops
import java.util.*;
public class NestedFor
{
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        System.out.println("Enter the number of rows");
        int n=s.nextInt();
        for(int i=1;i<=n;i++)//rows
        {
            for(int j=1;j<=i;j++)//processing of each rows
            {
                System.out.print(""+j);
            }
            System.out.println();//each row
        }
    }
}
/*
```

D:\Openelective2>javac NestedFor.java

D:\Openelective2>java -cp . NestedFor

Enter the number of rows

10

1

12

123

1234

12345

123456

1234567

12345678

123456789

12345678910

*/

```
//java program to demonstrate for-each
import java.util.*;
public class ForEach
{
    public static void main(String args[])
    {
        int a[]={10,20,30,40,50};
        for(int i: a)
        {
            System.out.println(i);
        }
    }
}
```

Jump Statements:

```
//java program to demonstrate Continue statement
import java.util.*;
public class Continue
{
    public static void main(String args[])
    {
        for(int i=1;i<=10;i++)
        {
            if(i==5)
            {
                continue;
            }
            System.out.println(i);
        }
    }
}
/*
```

D:\Openelective2>javac Continue.java

D:\Openelective2>java -cp . Continue

The Java ternary operator (or conditional operator) is a concise, single-line alternative to a simple if-else statement. It takes three operands: a condition, an expression to return if the condition is true, and an expression to return if the condition is false.

Syntax

The basic syntax is as follows:

java

variable = (Condition) ? Expression_If_True : Expression_If_False;

Condition : A boolean expression that evaluates to either true or false.

***Expression_If_True:** The value returned if the Condition is true.

***Expression_If_False:** The value returned if the Condition is false

//java program to implement greatest among three using ternary

```
import java.util.*;
public class Ternary
{
    public static void main(String args[])
    {
        Scanner s=new Scanner(System.in);
        System.out.println("Enter the three numbers");
        int a=s.nextInt();
        int b=s.nextInt();
        int c=s.nextInt();
        int result= (a>b&&a>c)?a:(b>c)?b:c;
        System.out.println(""+result+"is greater number");

    }
}
/*
```

D:\Openelective2>javac Ternary.java

D:\Openelective2>java -cp . Ternary

Enter the three numbers

16 1 24

24is greater number*/